Assignment No: 2

Git Branching and Collaboration for Distributed Software Development

Aim

To implement Git feature branching, pull requests, code review and merging to simulate collaborative development in a distributed team.

Output

1. Clone shared repository (Student B, collaborator)

```
aditya@aditya:~/devops-task-manager$ git clone https://github.com/aditya/devops-task-manager.git
Cloning into 'devops-task-manager'...
Repository cloned successfully.
aditya@aditya:~/devops-task-manager$ cd devops-task-manager
```

2. Synchronise main

```
aditya@aditya:~/devops-task-manager$ git switch main
Already on 'main'.
aditya@aditya:~/devops-task-manager$ git pull origin main
Main branch synchronized with remote repository.
```

3. Create feature branch

```
aditya@aditya:~/devops-task-manager$ git switch -c feature-add-task
Switched to a new branch 'feature-add-task'.
Task creation feature implemented.
aditya@aditya:~/devops-task-manager$ git branch
* feature-add-task
  main
```

4. Status, stage, commit

```
aditya@aditya:~/devops-task-manager$ git status
On branch feature-add-task
Changes not staged for commit:
        modified:   index.html
        modified:   script.js
aditya@aditya:~/devops-task-manager$ git add .
Changes staged successfully.
aditya@aditya:~/devops-task-manager$ git commit -m "Add task creation feature"
[feature-add-task 7d3a421] Add task creation feature
 2 files changed, 24 insertions(+)
```

5. Push feature branch

```
aditya@aditya:~/devops-task-manager$ git push -u origin feature-add-task
To https://github.com/aditya/devops-task-manager.git
 * [new branch]      feature-add-task -> feature-add-task
branch 'feature-add-task' set up to track 'origin/feature-add-task'.
Feature branch pushed successfully.
```

6. Pull Request on GitHub

```
Pull Request created:
  Title    : Add Task Creation Feature
  Base     : main
  Compare : feature-add-task
  Reviewer : Student B
feature-add-task -> main
Code review completed.
Pull Request approved.
Pull Request merged successfully.
```

7. Synchronise after merge

```
aditya@aditya:~/devops-task-manager$ git switch main
Switched to branch 'main'.
aditya@aditya:~/devops-task-manager$ git pull origin main
Updating 5c7b123..7d3a421
Fast-forward.
 index.html | 10 ++++++++--
 script.js | 14 ++++++++++++
Latest changes successfully pulled from the remote repository.
Add Task feature successfully integrated into the Student Task Manager application.
```

Conclusion

Feature branches, pull requests and code review were used to collaboratively add a feature and merge it into main, simulating distributed team development.